Practical : 5	**Build a small rule-based expert system or logic inference engine for a diagnosis or advisory problem.**
Date : 21/09/26	

Aim: -

To build a small rule-based expert system using Python to provide laptop troubleshooting advice based on user input.

Algorithm:-

1. Start the program.
2. Define a function named laptop_expert().
3. Accept three inputs: laptop power status, internet status, and laptop speed.
4. Check whether the laptop turns on.
5. If the laptop does not turn on, advise checking the power cable.
6. Otherwise, check whether the internet works.
7. If the internet does not work, advise checking Wi-Fi and restarting the router.
8. Otherwise, check whether the laptop is slow.
9. If the laptop is slow, advise closing unused programs and restarting the system.
10. If none of the conditions match, display that no common problem was found.
11. Display the advice.
12. Stop the program.

Program:-

```
# Practical 5: Simple Rule-Based Expert System
# Laptop Troubleshooting Expert System

def laptop_expert(turns_on, internet_works, is_slow):

    if turns_on == "no":
        return "Check power cable of the laptop"
```

```
    elif internet_works == "no":
        return "Check WiFi and restart the router"

    elif is_slow == "yes":
        return "Close unused programs and restart the system"

    else:
        return "No common problem in the expert system"


# Get input from the user
turns_on = input("Does the laptop turn on? yes/no: ").lower()

internet_works = input("Does the internet work? yes/no: ").lower()

is_slow = input("Is the laptop slow? yes/no: ").lower()


# Display the expert system advice
advice = laptop_expert(turns_on, internet_works, is_slow)

print("\nExpert System Advice")
print(advice)
```

Output:-

Laptop internet problem:-

```
C:\Users\Vishnu\OneDrive>python AI_Practical_5.py
Does the laptop turn on? yes/no: yes
Does the internet work? yes/no: yes
Is the laptop slow? yes/no: no

Expert System Advice
No common problem in the expert system
```

Laptop running slowly:-

```
C:\Users\Vishnu\OneDrive>python AI_Practical_5.py
Does the laptop turn on? yes/no: yes
Does the internet work? yes/no: yes
Is the laptop slow? yes/no: yes

Expert System Advice
Close unused programs and restart the system
```

Laptop power problem:-

```
C:\Users\Vishnu\OneDrive>python AI_Practical_5.py
Does the laptop turn on? yes/no: no
Does the internet work? yes/no: no
Is the laptop slow? yes/no: no

Expert System Advice
Check power cable of the laptop
```

Conclusion:-

Thus, a simple rule-based expert system was successfully implemented using Python. The system accepts user inputs, evaluates predefined IF-THEN rules, and provides suitable laptop troubleshooting advice. This practical demonstrates the application of rule-based reasoning in Artificial Intelligence.